7.30

We extracted genomic DNA from *Dunaliella salina* to amplify the endogenous P_{DCA1} promoter. We selected P_{DCA1} as a candidate regulatory element based on its reported salt-responsive activity in transformed *D. salina*. We tested direct amplification and a two-step primer strategy, and prepared a linearized pCAMBIA1302 backbone by PCR. We expected fragments of approximately 1,395 bp for the P_{DCA1} promoter and 10,549 bp for the backbone.

7.30 PCR targets

Target	Expected size	Recorded approach / missing details
P_{DCA1}	Approximately 1,395 bp	Direct and two-step primer strategies; reaction composition and cycling program not recorded
pCAMBIA1302 backbone	Approximately 10,549 bp	Linearized by PCR; reaction composition and cycling program not recorded

8.5

We planned PCR amplification of the safety-module fragments. We recorded expected sizes of mazF, 336 bp; P_{Pip}, 2,044 bp; mazE, 249 bp; and NOS, 253 bp. We listed KOD polymerase for mazF, P_{Pip}, and mazE; extension time was 20 s for P_{Pip} and approximately 10 s for the other two fragments. We planned to miniprep the EGFP template, amplify *EGFP* and P_{DCA1}, check and recover the products, assemble the fragments with the prepared backbone by Gibson assembly, transform DH5α, and pick colonies the next day.

Planned PCR fragments and expected sizes

Target fragment	Expected size (bp)	Recorded PCR note
mazF	336	KOD polymerase; approximately 10 s extension
P_{ip} promoter	2,044	KOD polymerase; 20 s extension
mazE	249	KOD polymerase; approximately 10 s extension

8.6

We miniprepped the EGFP template plasmid and amplified *EGFP* with homology-arm primers. We amplified the P_{DCA1} promoter in parallel, checked both products by agarose-gel electrophoresis, and recovered the target fragments. We recorded concentrations of 169 for P_{DCA1} and 340.6 for *EGFP*; the units were not included in the text record.

8.7

We assembled the recovered *EGFP* and P_{DCA1} promoter fragments with the linearized vector. We set up a Gibson reaction with p2301 (7 µL), EGFP (0.5 µL), P_{DCA1} promoter (0.5 µL), 2× MultiF Mix (10 µL), and ddH_2O (2 µL). We incubated the reaction at 50°C for 30 min and held it at 4°C.

Gibson assembly reaction

Component	Volume
p2301 linearized backbone	7 µL
EGFP fragment	0.5 µL
P_{DCA1} fragment	0.5 µL
2× MultiF Mix	10 µL
ddH_2O	2 µL

Assembly incubation

Step	Condition
Gibson assembly	50°C for 30 min; hold at 4°C

We added 10 µL of assembly product to 100 µL of Trelief 5α competent cells and incubated the mixture on ice for 15 min. We heat-shocked the cells at 42°C for 60 s, returned them to ice for 10 min, added 700 µL antibiotic-free LB, and recovered them at 37°C and 200 rpm for 45 min. We centrifuged the cells at 12,000 rpm for 1 min, resuspended them in 200 µL of supernatant, and spread them on kanamycin LB agar.

We observed few colonies on the plate. We did not determine the cause of the low colony count. We labelled this attempt as intended for p1302-P_{DCA1}-EGFP, but recorded p2301 as the assembly backbone. In a later sequencing entry, we identified the submitted p2301 sample as empty. We kept this early p2301 attempt separate from p1302-P_{DCA1}-mgfp5.

8.12

We amplified mazF with homology arms for a salt-responsive construct and amplified *EGFP* and P_{DCA1} promoter fragments in parallel. We considered designs for p1302-P_{DCA1}-mazF and p1302-P_{DCA1}-EGFP-mazF. We did not record a clear band result or product concentration in this entry.

PCR information recorded on 8.12

Target	Recorded information	Not recorded in this entry
mazF	Amplified with homology arms for Gibson assembly	Reaction composition and cycling program
EGFP and P_{DCA1} promoter	Amplified in parallel	Primer names, reaction composition, and cycling program

8.13

We re-amplified the P_{DCA1} promoter, *EGFP*, and mazF, checked the products by agarose-gel electrophoresis, recovered the target bands, and measured DNA concentrations. We recorded *EGFP* as 65.1, but did not record its unit or the other concentrations and sample identities in this entry.

PCR information recorded on 8.13

Target	Recorded outcome	Not recorded in this entry
P_{DCA1} promoter, EGFP, and mazF	Products checked on a gel and target bands recovered; *EGFP* concentration recorded as 65.1 without units	Reaction composition, cycling program, full concentrations, and sample identities

8.19

We amplified *mazF* by PCR for the CaMV 35S-mazF expression construct. We digested the empty pCAMBIA1302 backbone and *mazF* insert separately, recovered the products from a gel, and ligated them with T4 DNA ligase. We transformed *E. coli* with the ligation product, plated the cells on selective medium, and planned colony-PCR screening followed by sequencing of positive clones. We did not record the restriction enzymes or downstream screening results in this entry.

PCR and cloning details recorded on 8.19

Step	Recorded information	Not recorded in this entry
mazF PCR	*mazF* insert amplified by PCR	Reaction composition, primers, and cycling program
Restriction and ligation	Empty pCAMBIA1302 backbone and insert digested and ligated with T4 DNA ligase	Restriction enzymes and reaction volumes

8.26

We started the safety-module build and prepared the materials. We dissolved the listed primers (including dca1-F1, dca1-R, 2301-dca1-F, EGFP-R, mazF-F, and mazF-R) as 100 µmol/L stocks and diluted working stocks to 10 µmol/L. We streaked DH5α on LB agar for activation and streaked the p1302-CaMV 35S-mazF glycerol stock on LB agar containing kanamycin (50 µg/mL). We prepared LB media and modified PKS/DM medium containing 1.5 M NaCl. We inoculated 1 mL of the FACHB-435 algal stock into 50 mL PKS/DM (1.5 M NaCl) and cultured it at 26°C under 16 h light per day at 50 µmol m^{-2} s^{-1}.

Primer sequences recorded in the dated notes and supplied DNA maps

Primer name	Sequence (5′–3′)	Source / status
dca1-F1	atcttcgtcaacatggtggaCAGCAAGTAAAGCACTAGTTAAGGGC	Exact name in p2301-Pdca1-EGFP map
dca1-R	cccttagacatCTTGGGACACCGCAAATGG	Map annotation: dca1-R1
2301-dca1-F	ATTACAGGTGACCAGCTCGAATTT	Exact name in p2301-Pdca1-EGFP map
EGFP-R	tcgagctggtcacctgtaatTTACTTGTACAGCTCGTCCATGCC	Exact name in p2301-Pdca1-EGFP map
mazF-F	gaacacgggggactcttgacATGGTGTCTCGCTACGTGCC	Exact name in supplied mazF maps
mazF-R	gatcggggaaattcgagctgTTAACCAATCAGCACGTTAATCTTAGC	Exact name in supplied mazF maps
dca1-gibson-F	atcttcgtcaacatggtggaCAGCAAGTAAAGCACTAGTTAAGGGC	Map annotation: PDCA1-gibson-F
dca1-gibson-R	accatggtcaagagtccccCTTGGGACACCGCAAATGGG	Map annotation: PDCA1-gibson-R
Pip-F	Not located under this exact name in the supplied DNA maps	Named in the 9.28 colony-PCR note; sequence not assigned
ip-F	ACCACAAGGGAATGCTCAAATGC	Map annotation in the composite p1302 file; equivalence to Pip-F unverified
ip-R	CTTGACAAGATGCTTTTTGATGAAAGTGTTC	Map annotation in the composite p1302 file; use/date not linked in the note
ip-gibson-F	agatcACCACAAGGGAATGCTCAAATG	Map annotation in the composite p1302 file; use/date not linked in the note
ip-gibson-R	cgagacaccatCTTGACAAGATGCTTTTTGATGAAA	Map annotation in the composite p1302 file; use/date not linked in the note
p1302-dca1-F	tccaccatgttgacgaagattttct	Map annotation in supplied p1302 maps; use/date not linked in the note
1302-mazF-F	CAGCTCGAATTTCCCCGATC	Map annotation in p1302-CaMV35S-mazF file
1302-mazF-R	GTCAAGAGTCCCCCGTGTTC	Map annotation in p1302-CaMV35S-mazF file
Pdca1-F	CAGCAAGTAAAGCACTAGTTAAGGGC	Map annotation in composite p1302 file
Pdca1-R	CTTGGGACACCGCAAATGGG	Map annotation in composite p1302 file
DCA1-gfp-F	GCATCCTTGTGCAGCATCCC	Map annotation in composite p1302 file
1302-vector-F	CAGCTCGAATTTCCCCGATC	Map annotation in composite p1302 file
1302-vector-R	GTCAAGAGTCCCCCCTTGGG	Map annotation in composite p1302 file
mazE-F	cccaaggggggactcttgacATGATCCATAGTAGCGTGAAGCG	Map annotation in composite p1302 file
mazE-R	gaacgatcTTACCACACCTCCTTGTCCTTGG	Map annotation in composite p1302 file
terminator-F	gaggtgtggtaaGATCGTTCAAACATTTGGCAATAA	Map annotation in composite p1302 file
terminator-R	gcattcccttgtggtGATCTAGTAACATAGATGACACCGCG	Map annotation in composite p1302 file
mazF-ip-F	cttgtcaagATGGTGTCTCGCTACGTGCC	Map annotation in composite p1302 file
mazF-ip-R	gatcggggaaattcgagctgTTAACCAATCAGCACGTTAATCTTAGC	Map annotation in composite p1302 file
2301-dca1-R	TCCACCATGTTGACGAAGATTTTCT	Map annotation in p2301-Pdca1-EGFP file
EGFP-F	tgtcccaagATGTCTAAGGGCGAGGAGCTG	Map annotation in p2301-Pdca1-EGFP file

We observed single bacterial colonies the following day and described the algal culture as clear, with no visible contamination. We also found a separate Benchling entry titled “8.14” with a date field of 8.26, so we noted the discrepancy between its title and date field.

8.27

We amplified the P_{DCA1} promoter from *D. salina* genomic DNA for the p2301-P_{DCA1}-EGFP reporter attempt. We measured the genomic-DNA concentration and checked its integrity on a 0.8% agarose gel. We set up five PCR tubes: one for gel checking and four for product recovery. Each 50 µL reaction contained 2 µL genomic DNA, 2 µL dca1-F1 (10 µmol/L), 2 µL dca1-R (10 µmol/L), 25 µL 2× Phanta Ultra Master Mix, and 19 µL ddH_2O. We used the program 98°C for 3 min; 34 cycles of 98°C for 10 s, 58°C for 20 s, and 72°C for 45 s; then 72°C for 5 min. We observed a single band at the expected position and no band in the no-template control, then excised and recovered the target band.

P_{DCA1} PCR reaction (50 μL)

Component	Stock / description	Volume
Genomic DNA template	D. salina genomic DNA	2 μL
dca1-F1	10 μmol/L	2 μL
dca1-R	10 μmol/L	2 μL
2× Phanta Ultra Master Mix	2×	25 μL
ddH_2O	—	19 μL
Total		50 μL

P_{DCA1} PCR cycling program

Stage	Temperature	Time	Cycles
Initial denaturation	98°C	3 min	1
Denaturation	98°C	10 s	34
Annealing	58°C	20 s	34
Extension	72°C	45 s	34
Final extension	72°C	5 min	1

We ran a 1.5% agarose gel (0.9 g agarose in 60 mL 1× TAE with 6 μL GelRed) at 130 V for 40 min using a 100–2,000 bp marker. We observed a single clear band at the expected position and no band in the no-template control. We excised and recovered the target band for the subsequent amplification step.

8.28

We introduced vector-homology arms onto the recovered P_{DCA1} promoter fragment by a second PCR using dca1-gibson-F/R primers. We used a two-step amplification strategy to match the homology arms to the genomic template. We linearized p2301 by double digestion at a unique site. Our note lists the enzyme as “AfII”; we have not verified this spelling against the original plasmid map.

Second-round PDCA1 PCR

Parameter	Recorded information
Target	P_{DCA1} promoter fragment with vector-homology arms
Primers	dca1-gibson-F/R; the sequence map labels the pair PDCA1-gibson-F/R
Approach	Two-step amplification from the genomic template
Reaction composition and cycling program	Not recorded in the dated entry

We assembled the P_{DCA1} promoter insert and linearized p2301 by Gibson Assembly at an insert-to-vector molar ratio of approximately 3:1, using 50–100 ng vector and 10 μL 2× Gibson Master Mix in a 20 μL total reaction. We incubated the assembly at 50°C for 60 min. We transformed DH5α by incubating the cells on ice for 30 min, heat-shocking at 42°C for 90 s, and returning them to ice for 2 min. We added 500 μL LB, recovered the cells at 37°C for 1 h, and spread 200 μL on LB-kanamycin (50 μg/mL) plates. We observed single colonies the following day.

Gibson assembly setup

Component / parameter	Recorded amount or condition
Linearized p2301 backbone	50–100 ng; volume not recorded
P_{DCA1} promoter insert	Insert:vector molar ratio approximately 3:1; volume not recorded
2× Gibson Master Mix	10 μL
Total reaction volume	20 μL
Incubation	50°C for 60 min

8.29

We picked ten colonies from the transformation plate, resuspended each in 20 µL ddH_2O, and used 2 µL as template for colony PCR. We used 2301-dca1-F and EGFP-R to amplify across the P_{DCA1} promoter–EGFP junction. Each 20 µL reaction contained 2 µL colony suspension, 0.5 µL of each 10 µmol/L primer, 10 µL 2× Taq Master Mix, and 7 µL ddH_2O. We used PCR conditions of 95°C for 5 min; 30 cycles of 95°C for 30 s, 55°C for 30 s, and 72°C for 1 min; and 72°C for 5 min. We checked products on a 1% agarose gel at 90 V for 90 min. We observed the expected band in most colonies and selected two to three for overnight culture in 25 mL LB-kanamycin at 37°C and 220 rpm.

Colony PCR reaction (20 µL)

Component	Stock / description	Volume
Colony suspension template	—	2 µL
Forward primer	10 µmol/L	0.5 µL
Reverse primer	10 µmol/L	0.5 µL
2× Taq Master Mix	2×	10 µL
ddH_2O	—	7 µL
Total		20 µL

Colony PCR cycling program

Stage	Temperature	Time	Cycles
Initial denaturation	95°C	5 min	1
Denaturation	95°C	30 s	30
Annealing	55°C	30 s	30
Extension	72°C	1 min	30
Final extension	72°C	5 min	1

8.30

We miniprepped plasmids from the selected p2301-P_{DCA1}-EGFP colonies using a column kit and measured DNA concentration and purity. We checked the construct by double digestion and observed fragments matching the expected backbone and insert. We submitted sample 2301-P-E-1 for Sanger sequencing with primer 2301-dca1-F. In parallel, we inoculated a single p1302-CaMV 35S-mazF colony into LB-kanamycin for overnight culture and plasmid preparation.

8.31

We miniprepped p1302-CaMV 35S-mazF, measured its concentration, and checked the construct by double digestion. We observed release of an approximately 400 bp mazF fragment, consistent with the expected insert. We submitted the mazF region for sequencing and prepared a glycerol stock of the selected bacterial clone for storage at −80°C.

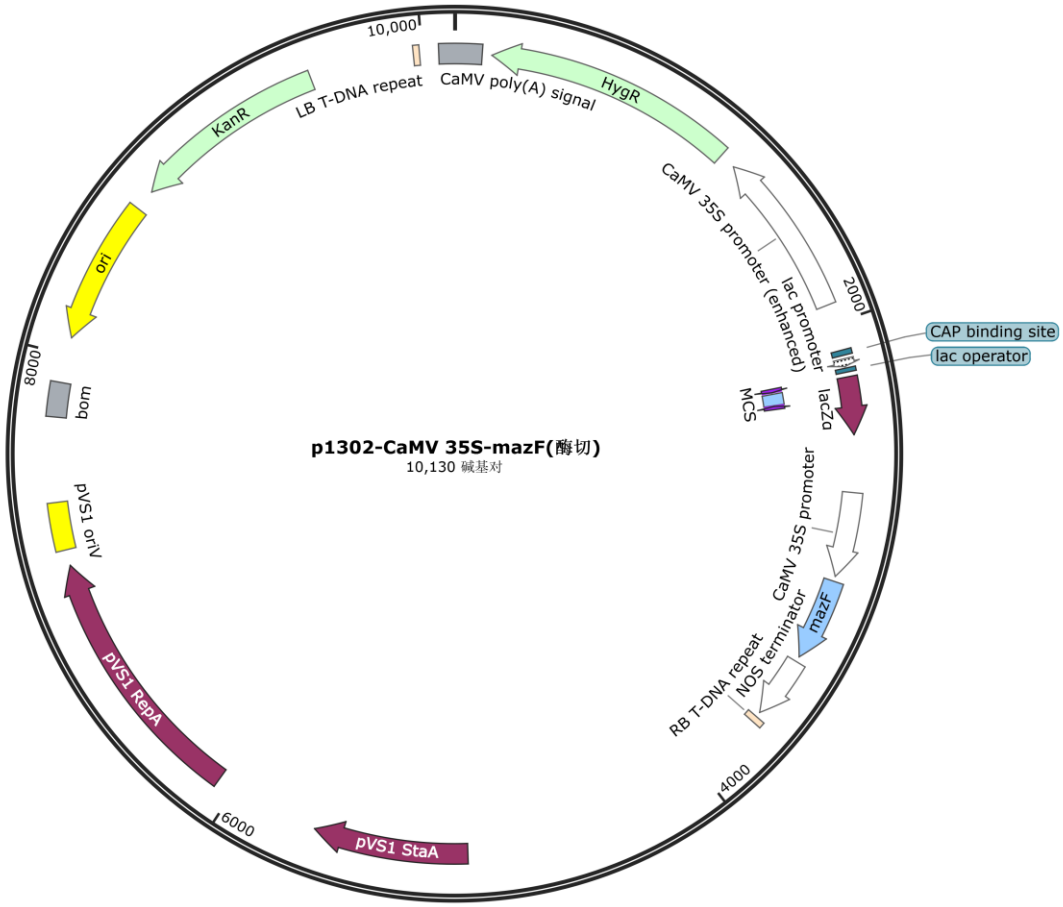


Figure 1. Restriction-map view supplied for p1302-CaMV35S-mazF.

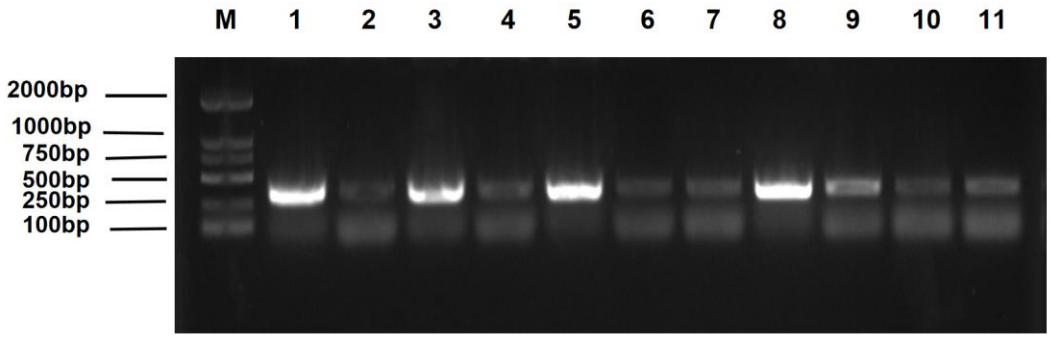


Figure 2. Colony PCR result for p1302-CaMV35S-mazF.

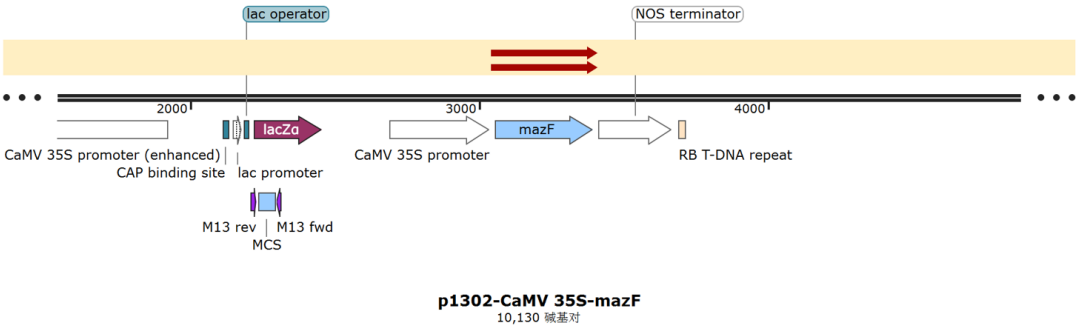


Figure 3. Sequence alignment result for p1302-CaMV35S-mazF.

9.1

We aligned the Sanger chromatograms (.ab1) to the design sequences. For p2301-P_{DCA1}-EGFP sample 2301-P-E-1, we obtained a 3,178 bp read with clear peaks, Q50 average quality, and 100% of bases at Q≥20. The read covered the T-DNA and P_{DCA1} promoter–EGFP junction and matched the design without a frameshift or unexpected variant. We confirmed the mazF region and CaMV 35S–mazF junction in p1302-CaMV 35S-mazF. We used these results to validate the p2301 reporter attempt and the p1302-CaMV 35S-mazF clone.

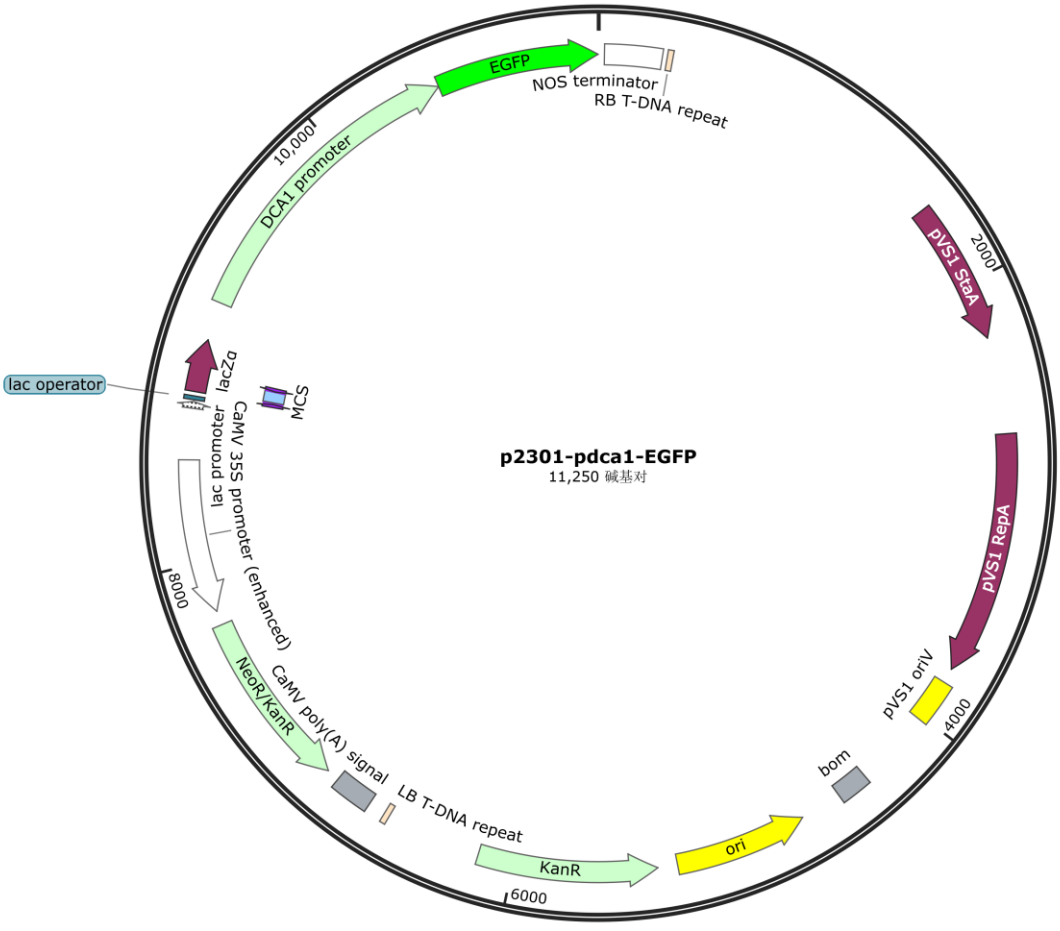


Figure 4. Circular plasmid map supplied for p2301-P_{DCA1}-EGFP.

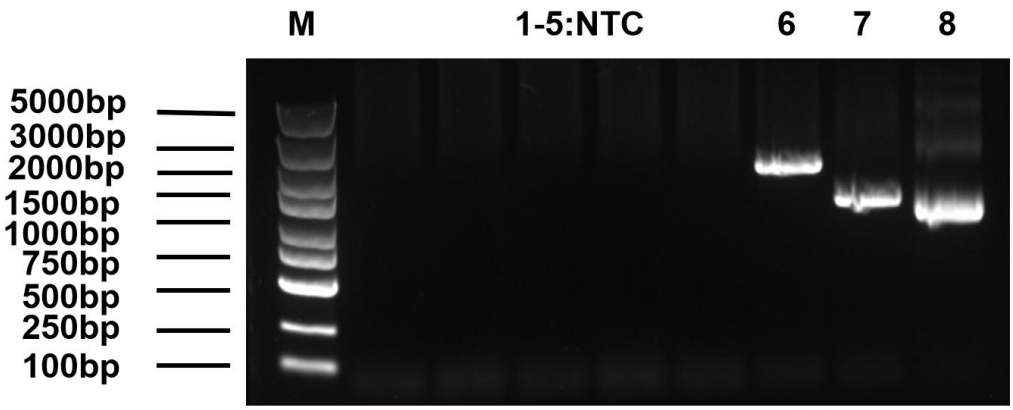


Figure 5. Colony PCR result supplied for p2301-P_{DCA1}-EGFP.

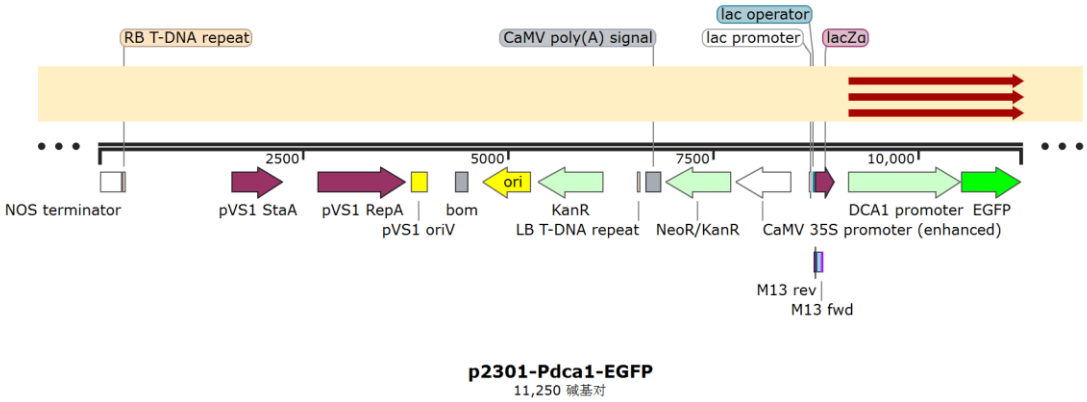


Figure 6. Sequence alignment result supplied for p2301-P_{DCA1}-EGFP.

We constructed p1302-P_{DCA1}-mgfp5 using the pCAMBIA1302 backbone. We screened recombinant colonies by colony PCR and confirmed the assembled sequence by sequencing.

9.2

We transferred the 8.26 algal starter culture into fresh PKS/DM containing 1.5 M NaCl at a 1:5–1:10 ratio and increased the culture volume from 100 mL to 500 mL. We maintained the cultures at 26°C under 16 h light per day at 50 μmol m^{-2} s^{-1}. We observed clear cultures and rising cell density.

9.3

We measured OD_{680} at the same time of day and checked cell morphology and culture clarity by microscopy. We observed a slight increase in OD_{680} compared with the previous day. We observed motile cells and no visible contamination.

9.4

We continued the algal expansion culture, measured OD_{680}, and replenished the culture with fresh medium as needed. We observed rising cell density and a deeper culture colour. We maintained the medium at 1.5 M NaCl.

9.5

We measured OD_{680} and estimated cell density by microscopy. We observed continued growth as the culture entered the exponential-growth phase. We kept cell density below the stationary-phase range while preparing for transformation.

9.6

We measured OD_{680} and expanded the culture volume where needed. We observed normal growth and no visible contamination. We transferred the cultures gently to limit mechanical damage to the algal cells.

9.7

We measured OD_{680} and checked cell morphology by microscopy. We observed increasing cell density as the culture approached the target range for transformation.

9.8

We measured OD_{680} and cell density. We observed that OD_{680} was within the recorded exponential-phase range of 0.5–1.0 and that cells appeared normal, with no visible contamination. We selected this culture for the upcoming transformation and genomic-DNA preparation.

9.9

We collected approximately 50 mL of exponential-phase algal culture by centrifugation (approximately 4,000 rpm for 5 min) and extracted genomic DNA using a column-based plant DNA kit. We checked DNA integrity on a 0.8% agarose gel and measured concentration and purity by Nanodrop. We observed an intact genomic-DNA band that we considered suitable for PCR.

9.10

We performed the first osmotic-transformation attempt using p1302-CaMV 35S-mazF to test the construct in *D. salina*. We collected exponential-phase cells (OD_{680} 0.5–1.0), washed them three times in fresh PKS/DM, adjusted the suspension to 10^6 cells/mL, and pre-adapted the cells in 1 M NaCl medium for 8 h. We pelleted 1 mL of cells and resuspended them in 1 mL PKS/DM containing 0.1 M NaCl. We added 5.46 μg p1302-CaMV 35S-mazF and 15 μL of 0.1% Triton X-100, vortexed vigorously for 90 s, and immediately transferred the mixture into 0.1 M NaCl recovery medium. We recovered the cells at 26°C under the recorded 16 h light schedule for 48 h. A no-plasmid treatment control was included.

We observed a clear culture and no obvious recovery growth during the first 48 h. The matched no-plasmid control also lost viability, so this first attempt did not isolate plasmid-specific toxicity as the cause.

9.11

At 24 h after the first transformation, we examined cell morphology and measured OD_{680}. We observed fewer cells than at the start, and some cells appeared shrunken or ruptured. We did not observe cell division. OD_{680} was below its initial value.

In parallel, we set up five PCR reactions for the P_{DCA1} promoter using the 9.9 *D. salina* genomic-DNA preparation: one reaction for gel checking and four for product recovery. We intended to use the amplified fragment for p1302-P_{DCA1}-mgfp5. We did not record the resulting gel, assembly, or sequencing details in this entry.

P_{DCA1} PCR setup recorded on 9.11

Parameter	Recorded information
Template	9.9 D. salina genomic-DNA preparation
Number of reactions	Five: one for gel checking and four for product recovery
Intended construct	p1302-P_{DCA1}-mgfp5
Reaction composition and cycling program	Not recorded in this entry
Gel, assembly, and sequencing outcome	Not recorded in this entry

9.12

At 48 h, we examined the transformed culture and no-plasmid treatment control by microscopy and measured OD_{680}. We observed few cells in the 0.1 M NaCl recovery condition and no recovery growth in either culture. We recorded the first transformation as unsuccessful. Because the no-plasmid control showed the same loss of viability, we could not attribute the shared effect to the mazF plasmid alone.

9.13

We reviewed the first-transformation outcome and revised the recovery condition. We considered the 0.1 M NaCl recovery medium a severe osmotic change for cells grown at high salt; Triton X-100 treatment and vigorous mixing added further stress. We therefore planned a second transformation with recovery medium containing 0.5 M NaCl, retained the 90 s mixing step, and included untreated, Triton-treated/no-plasmid, and empty-vector controls. We planned observations at 12, 24, and 48 h.

9.14

We repeated the pre-adaptation and cell-washing procedure and transformed parallel MazF and empty-vector (EV) groups. For each 1 mL cell suspension, we added either 5.46 μg p1302-CaMV35S-mazF or 5.46 μg p1302 empty vector, together with 15 μL 0.1% Triton X-100; the no-plasmid treatment control received Triton X-100 only. We vortexed each mixture for 90 s and immediately transferred it into 0.5 M NaCl PKS/DM recovery medium. We recovered the cultures at 26°C under the recorded light schedule for 48 h. We observed that all three groups remained viable and began to recover in density.

9.15

At 12 and 24 h after the second transformation, we examined cell morphology and measured OD_{680}. We observed that the MazF, EV, and no-plasmid groups remained viable and morphologically normal, with no obvious cell loss.

9.16

At 48 h, we observed recovery in cell density. We transferred the MazF and EV groups into 0.5 M NaCl PKS/DM liquid medium containing hygromycin B for enrichment and maintained a wild-type plus hygromycin control. We selected the working antibiotic concentration based on the team's minimum-inhibitory-concentration pretest, but did not record the concentration in this entry.

9.17

We inspected the liquid-selection cultures and checked cell abundance by microscopy. We observed viable MazF and EV cultures, while cell numbers in the wild-type plus hygromycin control began to decline.

9.18

We continued hygromycin enrichment and monitored culture density. We observed maintained or increasing density in the MazF and EV groups and a decline in the wild-type control. We replenished the cultures with fresh selective medium as scheduled.

9.19

We inspected the cultures microscopically and measured OD_{680}. We observed increasing density in the selected MazF and EV cultures, while the wild-type plus hygromycin control had almost no visible cells.

9.20

We replenished the selective cultures with fresh medium and checked for bacterial or fungal contamination. The enrichment cultures appeared uncontaminated and continued to grow.

9.21

We continued culture observation and measured OD_{680}. We observed viable MazF and EV cultures with increasing density under hygromycin selection.

9.22

We completed the liquid-enrichment stage. We observed survival of the MazF and EV cultures under hygromycin selection and loss of the wild-type control. We proceeded to isolate algal colonies and test for the mazF target sequence.

9.23

We serially diluted the enriched MazF culture (10^{-1}–10^{-4}) and spread it on solid PKS/DM containing 0.5 M NaCl, 2% agar, and hygromycin B to obtain individual algal colonies. We also collected 10 mL of enriched cells for genomic-DNA extraction. We set up PCR with mazF-F/mazF-R primers using transformed-cell genomic DNA and wild-type genomic DNA as the negative control. We expected an approximately 400 bp mazF amplicon and planned to analyse the gel the following day.

mazF genomic-DNA PCR record

Parameter	Recorded information
Primers	mazF-F and mazF-R
Template	Transformed-cell genomic DNA; wild-type genomic DNA as negative control
Expected amplicon	Approximately 400 bp
Reaction composition and cycling program	Not recorded in the dated entry

9.24

We analysed the genomic-DNA PCR products on a 1.5% agarose gel (0.9 g agarose in 60 mL 1× TAE with 4 µL GelRed) at 130 V for 40 min using a 5,000 bp marker. We observed an approximately 400 bp band in the transformed-cell DNA sample and no band in the wild-type control. We interpreted this result as detection of the mazF target sequence in the tested DNA preparation. We did not use this internal-primer PCR result alone to claim genomic integration. We observed algal colonies beginning to appear on the selective plates.

9.25

We picked individual algal colonies from the selective plates and inoculated at least three MazF and three EV lines separately into liquid PKS/DM containing 0.5 M NaCl. We expanded wild-type cells in parallel under the same conditions (26°C, 16 h light per day). We observed green cultures and increasing cell density.

9.26

We adjusted the WT, EV, and MazF cultures with fresh 0.5 M NaCl PKS/DM to a common starting OD_{680} of approximately 0.05. We used three separate culture vessels per group and maintained them at 26°C, 50 µmol m^{-2} s^{-1}, and 16 h light/8 h dark. We scheduled daily measurements from 9.27 through 10.07 (Day 0–Day 10).

In parallel, we amplified the P_{ip} promoter and mazF insert for p1302-P_{ip}-mazF assembly and prepared the vector backbone. Each 50 µL PCR contained 2 µL template, 2 µL each forward and reverse primer (10 µmol/L), 25 µL 2× KOD Master Mix, and 19 µL ddH_2O. We used the mazF program 98°C for 2 min; 34 cycles of 98°C for 10 s, 55°C for 30 s, and 68°C for 15 s; and 68°C for 2 min. Our first P_{ip}/backbone gel run was compromised when sample leaked from a damaged well. We repeated the analysis using 1.5% agarose for mazF and 1% agarose for P_{ip} and the backbone. We expected fragments of approximately 2 kb for P_{Pip}, 300–400 bp for mazF, and 10–11 kb for the backbone.

P_{ip}-mazF construct PCR reaction (50 µL per tube)

Component	Stock / description	Volume
Template	—	2 µL
Forward primer	10 µmol/L	2 µL
Reverse primer	10 µmol/L	2 µL
2× KOD Master Mix	2×	25 µL
ddH_2O	—	19 µL
Total		50 µL

mazF PCR cycling program

Stage	Temperature	Time	Cycles
Initial denaturation	98°C	2 min	1
Denaturation	98°C	10 s	34
Annealing	55°C	30 s	34
Extension	68°C	15 s	34
Final extension	68°C	2 min	1

Expected fragments and gel conditions

Fragment	Expected size	Gel concentration
P_{ip}	Approximately 2 kb	1% agarose
mazF	Approximately 300–400 bp	1.5% agarose
Vector backbone	Approximately 10–11 kb	1% agarose

9.27

We measured OD_{680} for the nine growth-curve cultures at Day 0 and recorded all replicate values. Mean ± SEM (n=3 cultures per group) was WT 0.050 ± 0.002; EV 0.048 ± 0.002; MazF 0.049 ± 0.001. We recorded culture colour, cell morphology, and visible contamination status.

In parallel, we recovered the P_{ip}, *mazF*, and backbone fragments from the gel. We measured concentrations of 55.05 ng/µL for the backbone, 298.15 ng/µL for P_{Pip}, and 304.25 ng/µL for mazF. We combined the fragments by Gibson assembly in a 10 µL reaction containing 5 µL 2× assembly mix, 3.6 µL backbone solution, and 1.4 µL P_{ip}/*mazF* fragment mixture. We incubated the reaction at 50°C for 30 min and held it at 4°C before transformation.

Gibson assembly reaction (10 µL)

Component	Volume / amount
2× assembly mix	5 µL
Recovered backbone solution	3.6 µL; 55.05 ng/µL
Pip/*mazF* fragment mixture	1.4 µL; individual Pip and mazF volumes not recorded
Incubation	50°C for 30 min; hold at 4°C

9.28

We measured OD_{680} for the nine cultures at Day 1 and recorded all replicate values. Mean ± SEM (n=3 cultures per group) was WT 0.067 ± 0.005; EV 0.054 ± 0.004; MazF 0.054 ± 0.003.

We transformed 5 µL of the P_{ip}-mazF assembly into DH5α competent cells, recovered the cells in antibiotic-free LB, and plated them on LB-kanamycin at 50 or 100 µg/mL. We picked eight colonies, boiled each colony suspension at 95°C for 10 min, and screened the lysates by colony PCR using Pip-F and mazF-R. We analysed the products on a 1% agarose gel with an empty-vector control in lane 9. We observed the expected bands in colonies 1, 2, 3, 6, and 8 and selected them for further culture. We did not find a final sequencing trace for p1302-Pip-mazF in the materials reviewed. The supplied map file is labelled “bar”; we have not assigned this map to the final pCAMBIA1302 construct until the label is reconciled.

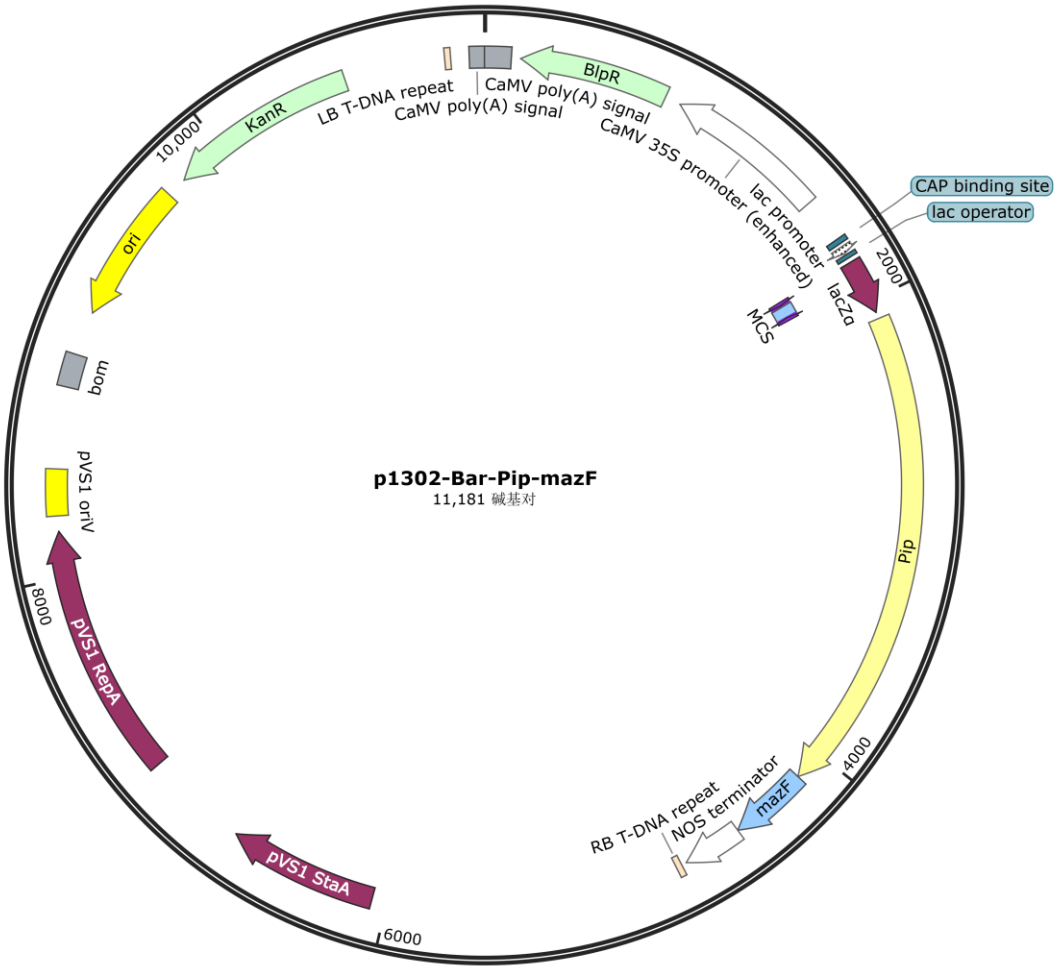


Figure .7 Circular plasmid map supplied for p1302-Bar-P_{ip}-mazF.

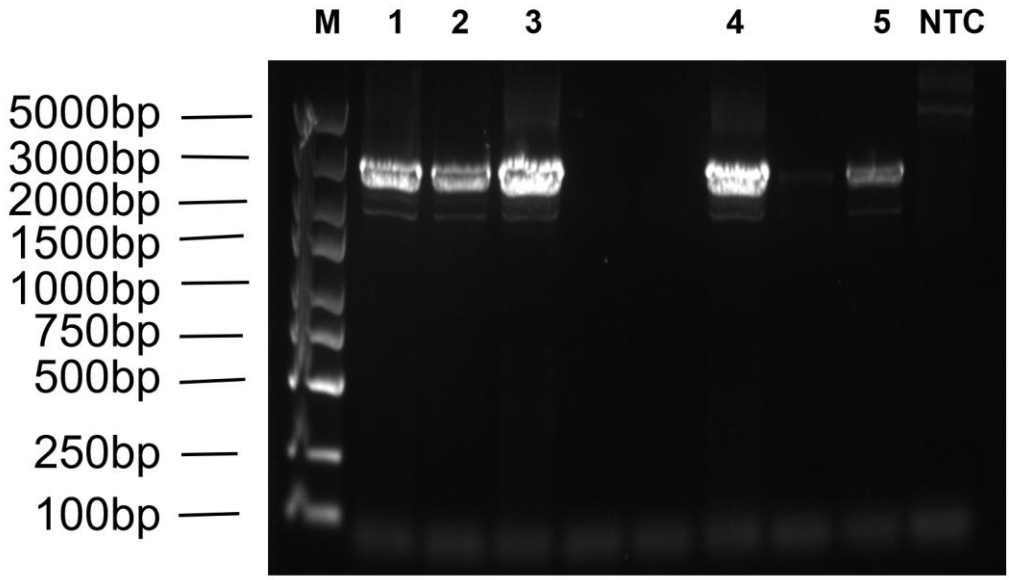


Figure 8. Colony PCR result supplied for p1302-Bar-P_{ip}-mazF.

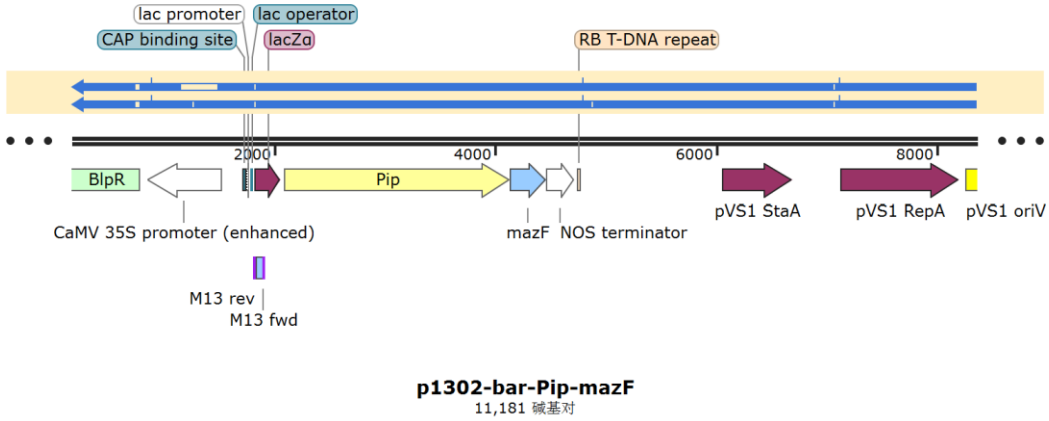


Figure 9. Sequence alignment result supplied for p1302-P_{ip}-mazF.

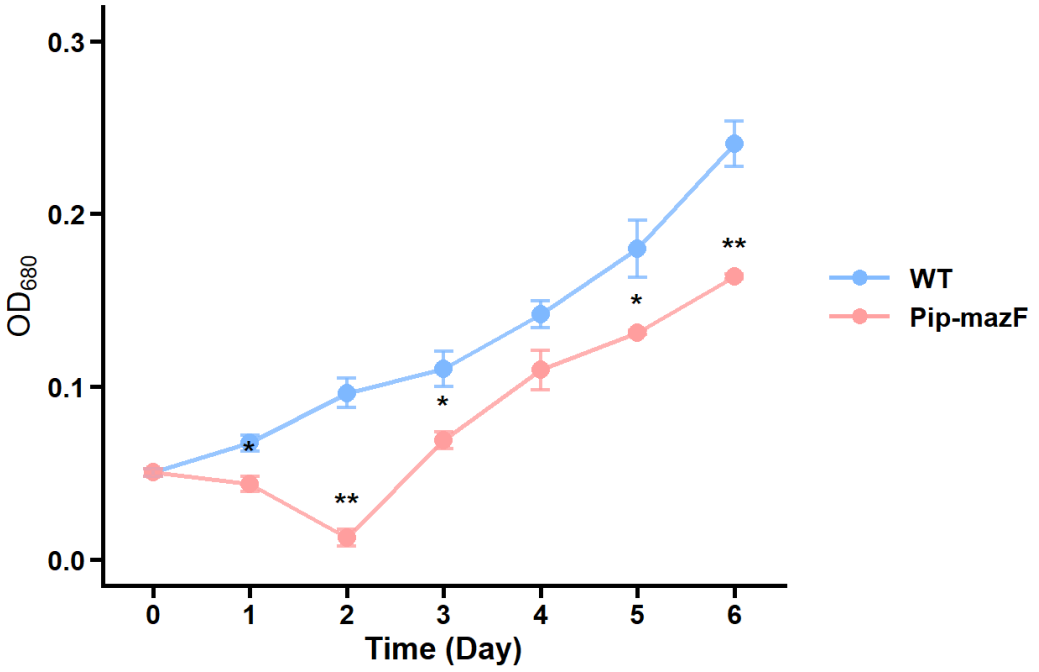


Figure 10. P_{ip}-mazF growth curve supplied with the project files; no matching dated measurements were recorded in this notebook.

Colony-PCR screening record

Parameter	Recorded information
Primer pair	Pip-F and mazF-R
Template	Boiled colony lysates from eight colonies
Reaction composition	Not recorded in the dated entry
Cycling program	Not recorded in the dated entry
Gel	1% agarose; empty-vector control in lane 9
Colonies selected for follow-up	1, 2, 3, 6, and 8

9.29

We measured OD_{680} for the nine cultures at Day 2 and recorded all replicate values. Mean ± SEM (n=3 cultures per group) was WT 0.096 ± 0.008; EV 0.068 ± 0.004; MazF 0.059 ± 0.003. We also checked colour, cell morphology, and visible contamination.

9.30

We measured OD_{680} for the nine cultures at Day 3 and recorded all replicate values. Mean ± SEM (n=3 cultures per group) was WT 0.110 ± 0.010; EV 0.094 ± 0.004; MazF 0.065 ± 0.003. We also checked colour, cell morphology, and visible contamination.

10.1

We measured OD_{680} for the nine cultures at Day 4 and recorded all replicate values. Mean ± SEM (n=3 cultures per group) was WT 0.142 ± 0.008; EV 0.112 ± 0.009; MazF 0.070 ± 0.003. We also checked colour, cell morphology, and visible contamination.

10.2

We measured OD_{680} for the nine cultures at Day 5 and recorded all replicate values. Mean ± SEM (n=3 cultures per group) was WT 0.180 ± 0.017; EV 0.141 ± 0.007; MazF 0.075 ± 0.003. We also checked colour, cell morphology, and visible contamination.

10.3

We measured OD_{680} for the nine cultures at Day 6 and recorded all replicate values. Mean ± SEM (n=3 cultures per group) was WT 0.241 ± 0.013; EV 0.165 ± 0.013; MazF 0.077 ± 0.003. We also checked colour, cell morphology, and visible contamination.

10.4

We measured OD_{680} for the nine cultures at Day 7 and recorded all replicate values. Mean ± SEM (n=3 cultures per group) was WT 0.269 ± 0.018; EV 0.205 ± 0.012; MazF 0.074 ± 0.003. We also checked colour, cell morphology, and visible contamination.

10.5

We measured OD_{680} for the nine cultures at Day 8 and recorded all replicate values. Mean ± SEM (n=3 cultures per group) was WT 0.329 ± 0.024; EV 0.219 ± 0.023; MazF 0.069 ± 0.004. We also checked colour, cell morphology, and visible contamination.

10.6

We measured OD_{680} for the nine cultures at Day 9 and recorded all replicate values. Mean ± SEM (n=3 cultures per group) was WT 0.346 ± 0.037; EV 0.266 ± 0.022; MazF 0.064 ± 0.004. We also checked colour, cell morphology, and visible contamination.

10.7

We completed the final OD_{680} measurement at Day 10 and compiled 11 time points from Day 0 through Day 10, representing 10 elapsed days. Day 10 mean ± SEM (n=3 cultures per group) was WT 0.360 ± 0.019; EV 0.281 ± 0.018; and MazF 0.059 ± 0.004. We calculated means and SEMs from the three replicate measurements at each time point and retained all 99 replicate-level observations in the accompanying CSV file.

We observed increasing mean OD_{680} in the WT and EV cultures. The MazF group reached a mean of approximately 0.077 at Day 6 and then declined to 0.059 at Day 10. We report these changes descriptively; we did not complete an inferential test and therefore do not report p-values or statistical significance. We did not infer cell viability or lysis from OD_{680} alone.

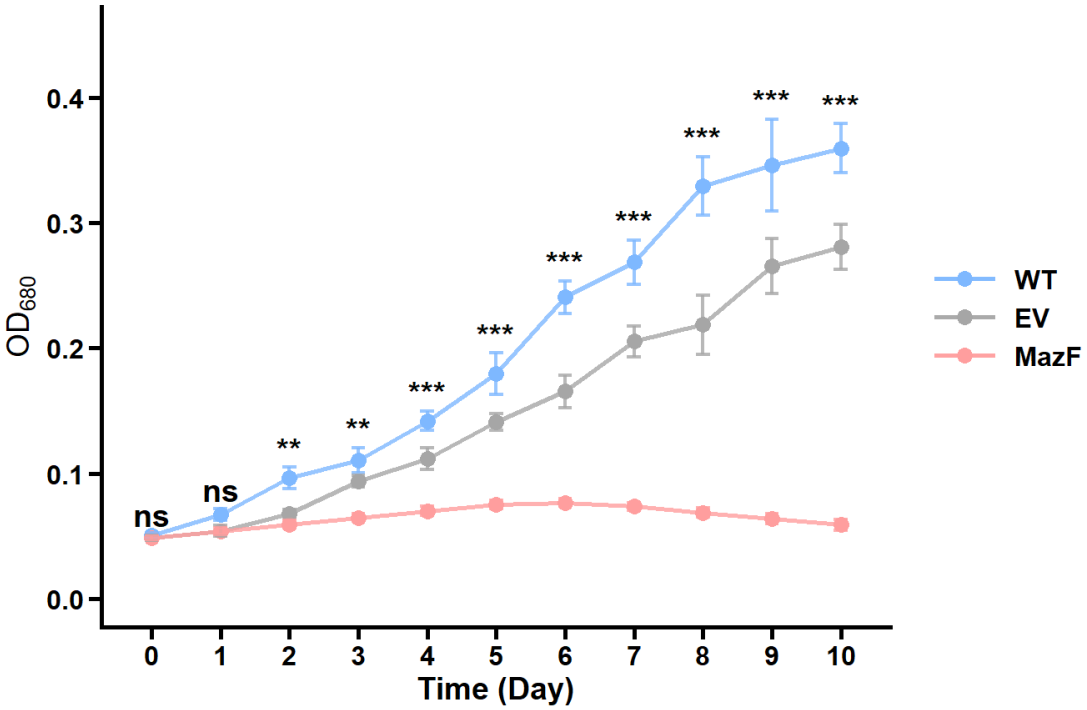


Figure 11. Growth curve of WT, empty-vector (EV), and MazF groups.